Diffusion Model
&
Guidance



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Content

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- Introduction
- Background

Caleb

- Motivation
- Methodology

Sam

- Experiments
- Conclusion

Chris

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Introduction

- **Diffusion models:** generate high-quality samples
- **Problem:** controlling the generation is challenging
- **Goal:** generate a specific classes/satisfy certain preferences
- **Solution:** guided generation (i.e. density-ratio guidance)

Background: Diffusion model

- **Forward process:** Adding noise to a real image until it becomes pure noise
- **Backward process:** Train a neural network to gradually remove the noise

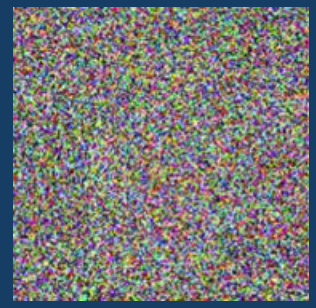
Forward process





Backward process





Training

- Train the neural network to minimize the average error between the actual and predicted noise

$$x_t = \sqrt{\overline{\alpha_t}}\, x_0 + \sqrt{1 - \overline{\alpha_t}}\epsilon$$

Clean image **Added noise**

$$\min_{\theta} E_{x_0,\epsilon}\left[\left|\left|\epsilon - \epsilon_\theta(x_t, t)\right|\right|^2\right]$$

Predicted noise

Sampling

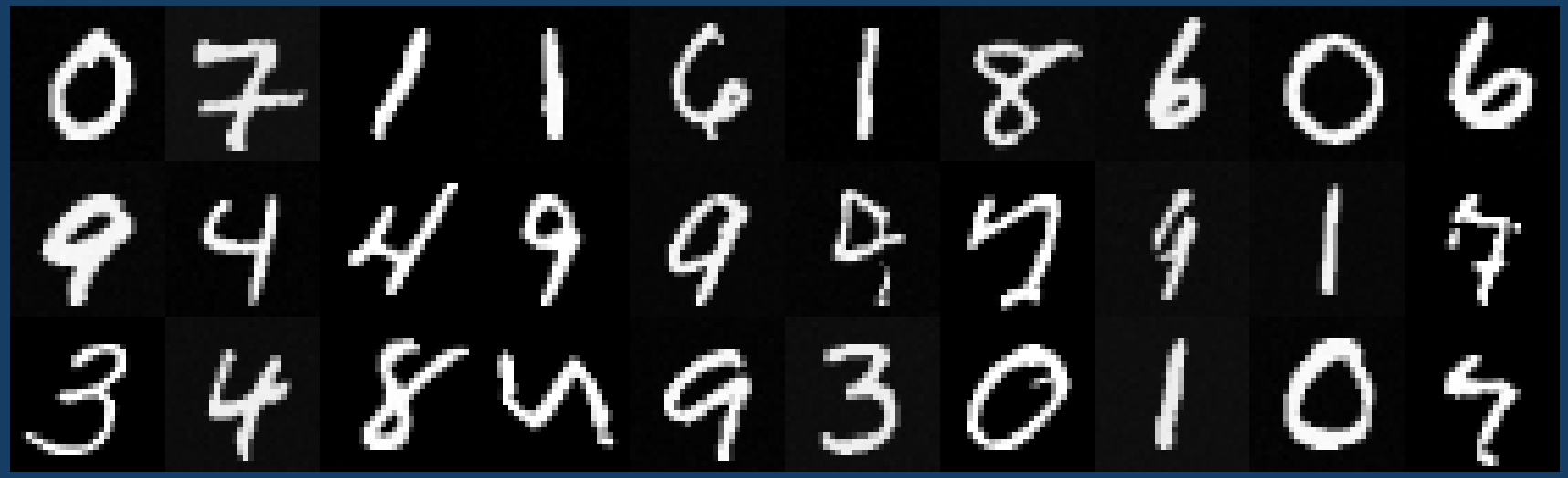
- Using the trained model, we can sample from the data distribution

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z$$

Trained Model

Diffusion Model Sampling

MNIST Dataset



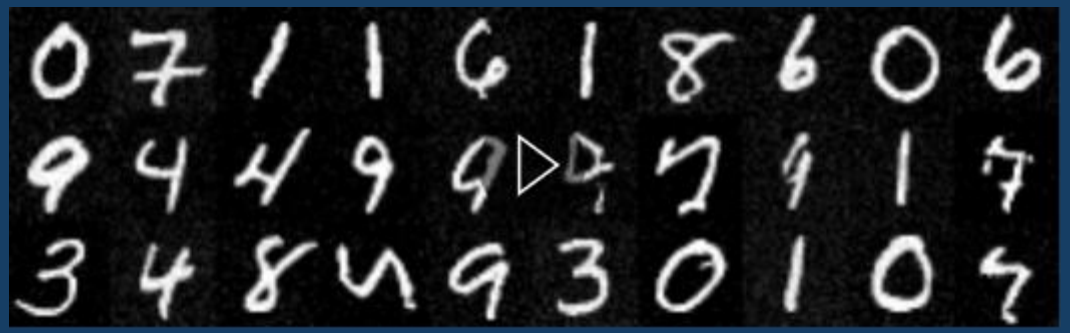
Motivation

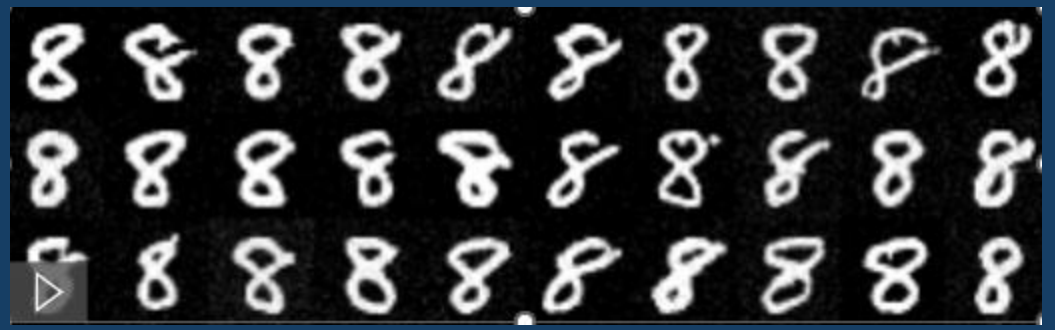
Limitation:

- Controllability: Unconditional models cannot generate samples with specific classes (e.g. number 8) or properties within the distribution they were trained on

Solution:

- Guided generation by modifying sampling without retraining the model





Methodology: Guidance

Main Idea: Guided generation by modifying sampling without retraining the model

The bridge:

Diffusion model (DDPM) sampling term $x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z$

+

Guidance term $\nabla_{x_t} \log\left(\frac{p}{q}\right)$

=

Guided sampling formula $x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z + \beta_t \nabla_x \log\left(\frac{p}{q}\right)$

DDPM sampling term **guidance term**

Mathematical derivation **:**

Song et al. (2021) in Appendix 3

Methodology: Density Ratio Guidance

Structure:

Density ratio: $r = \frac{p}{q}$ **p:** target distribution **q:** non-target distribution

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z + g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term **guidance term**

Goal: Estimate by logistic regression (MLP)

$$f_\theta(x) = \log r(x) = \log \frac{p(x)}{q(x)}$$

Method: logistic regression (MLP)

Training objective:

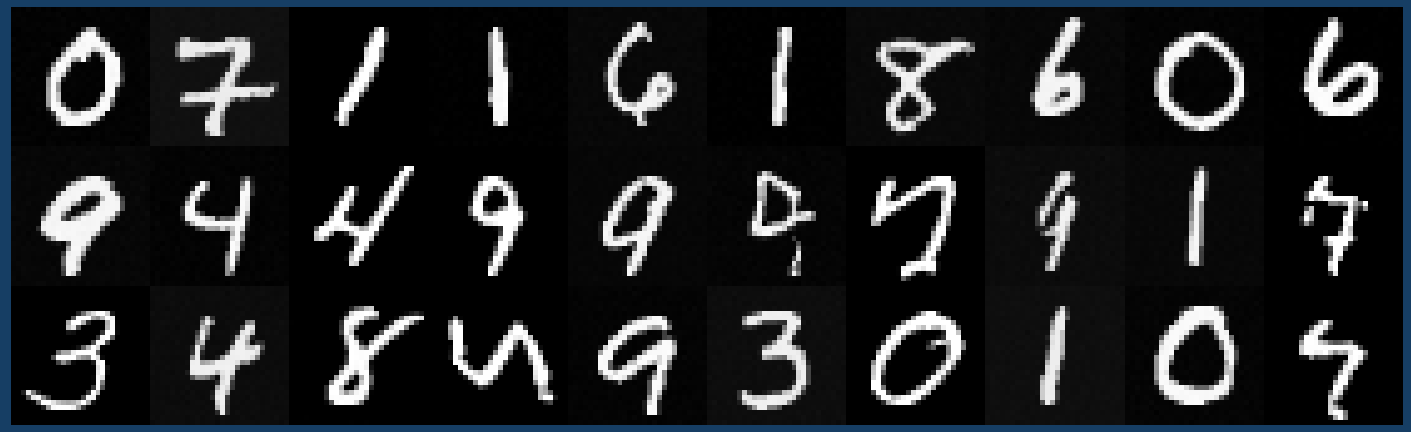
$$Loss = \text{BCEWithLogitsLoss}(f_\theta(x), y)$$

Guidance:

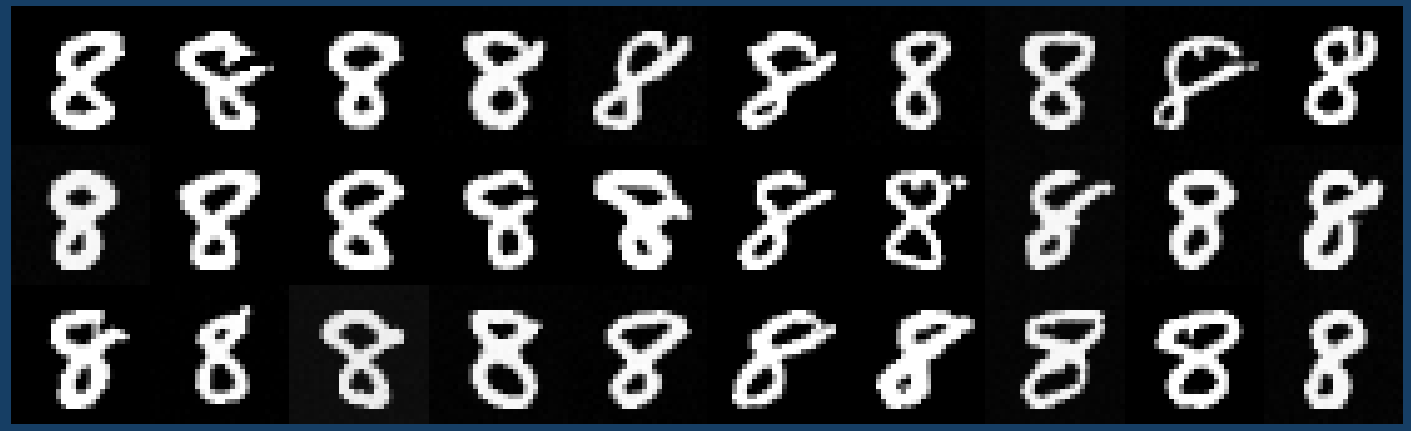
$$x_{t-1} = \text{DDPM step} + g \cdot \beta_t \nabla_x \log r(x)$$

Diffusion Model Sampling

Without guidance



Density ratio guidance to 8



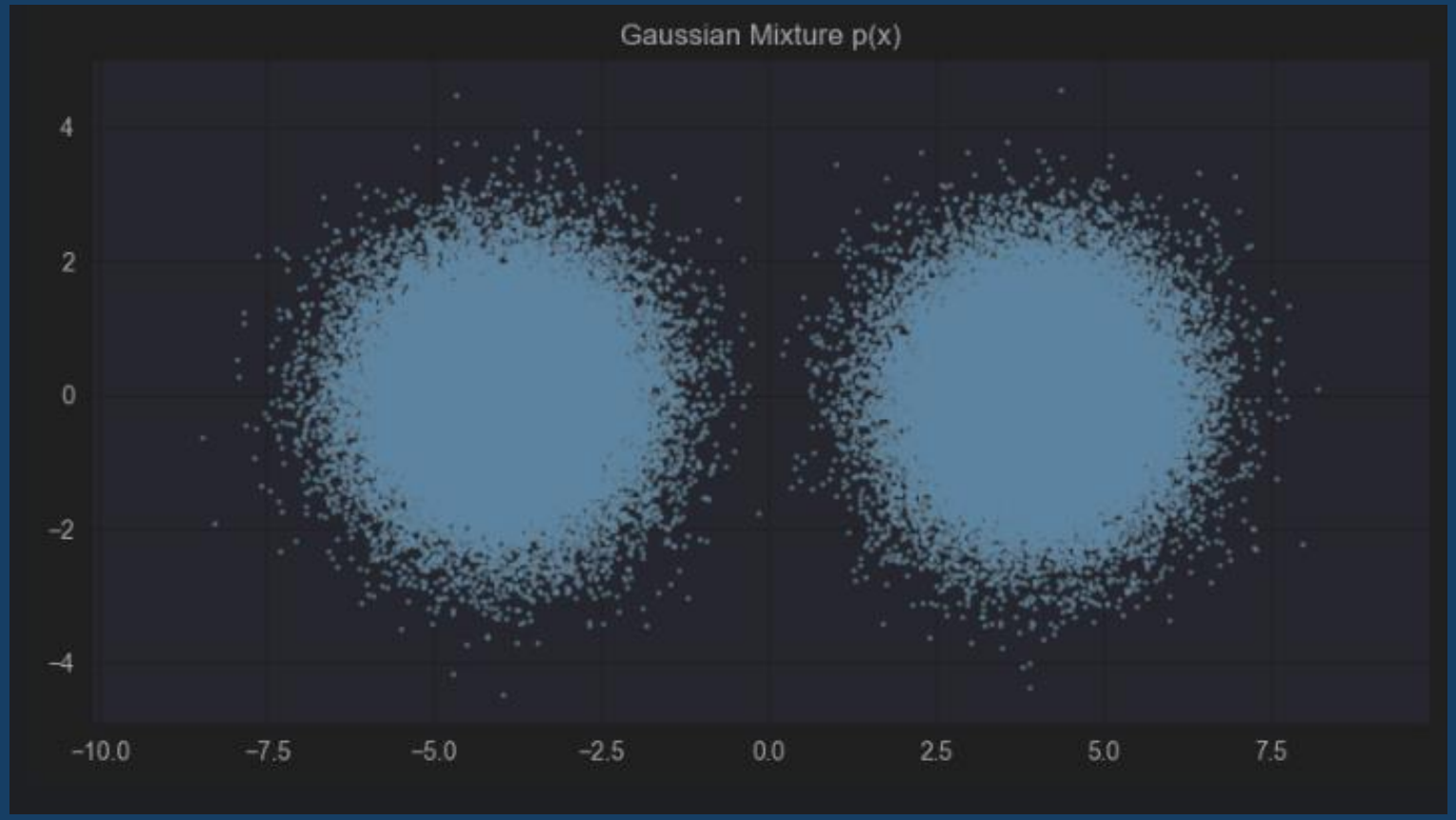
Experiments

Experiment 1: Toy dataset (2D Gaussian)

Experiment 2: Image dataset (MNIST numbers)

Experiments

Toy dataset (2D Gaussian)



Experiment Setting

- **Dataset:** 2D Gaussian mixture
- **Sampling:** Density ratio guidance

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z + g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term | guidance term

- **Quantitative Metrics**

Training side

Average loss →

Sampling side

Coverage (real→gen): how many real data are covered by generated data

Coverage (gen→real): how many generated data fall inside real distribution

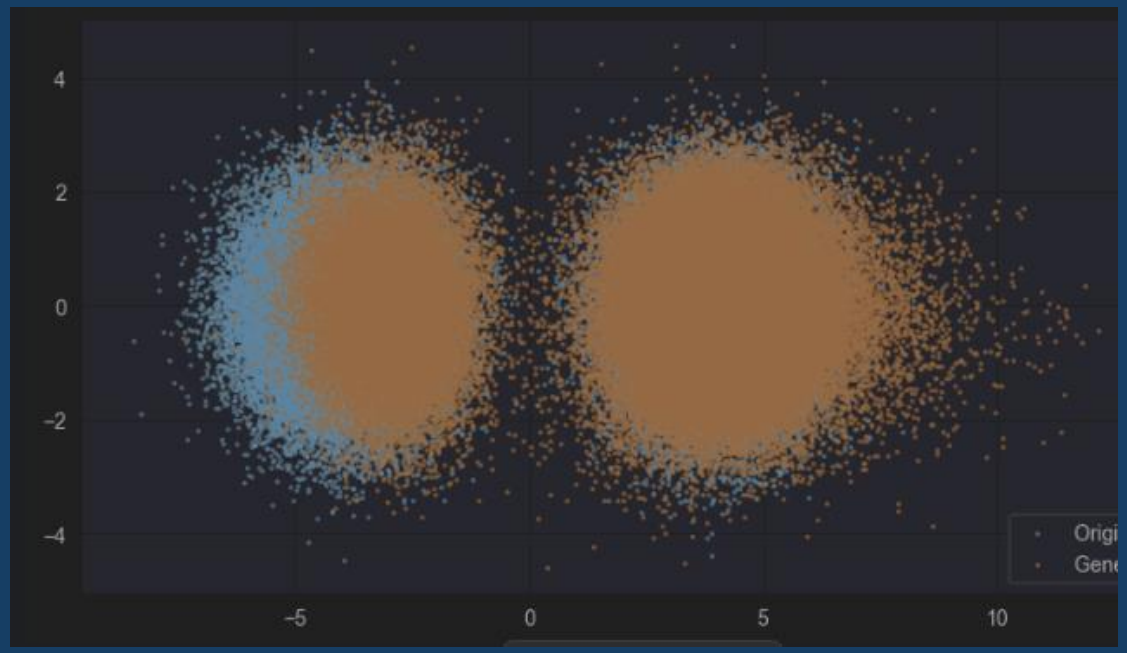
MMD²: The difference between the generated and real distribution

Higher, better ↑

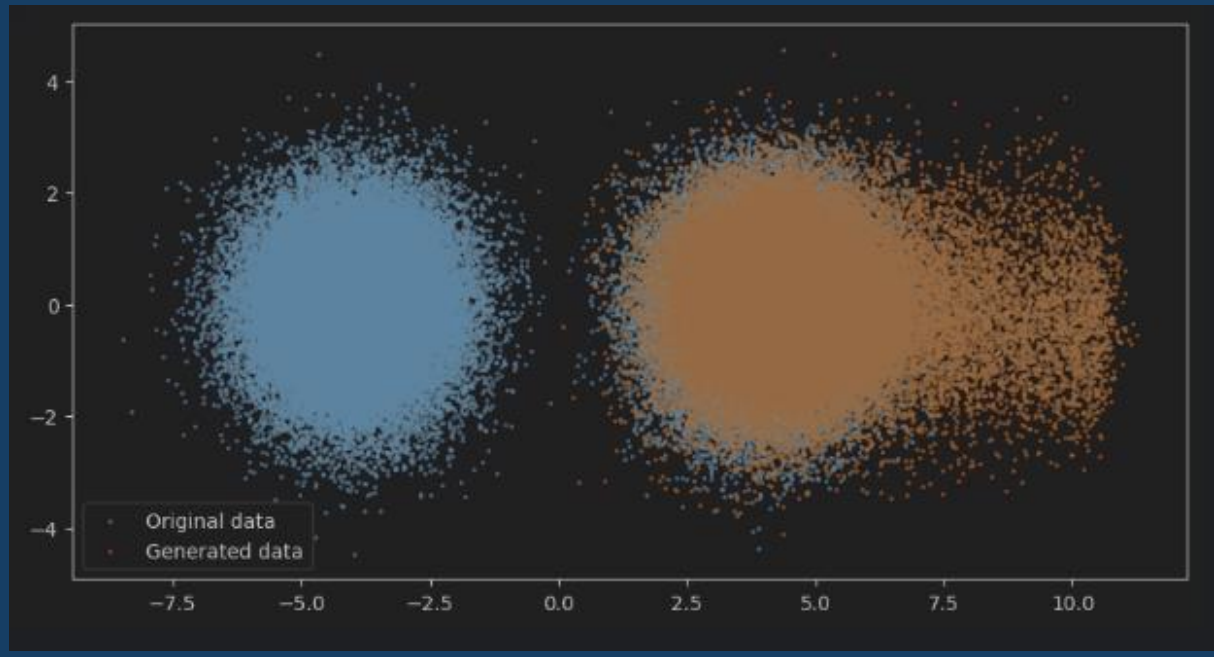
↓ Lower, better

Experiment Results

Without guidance



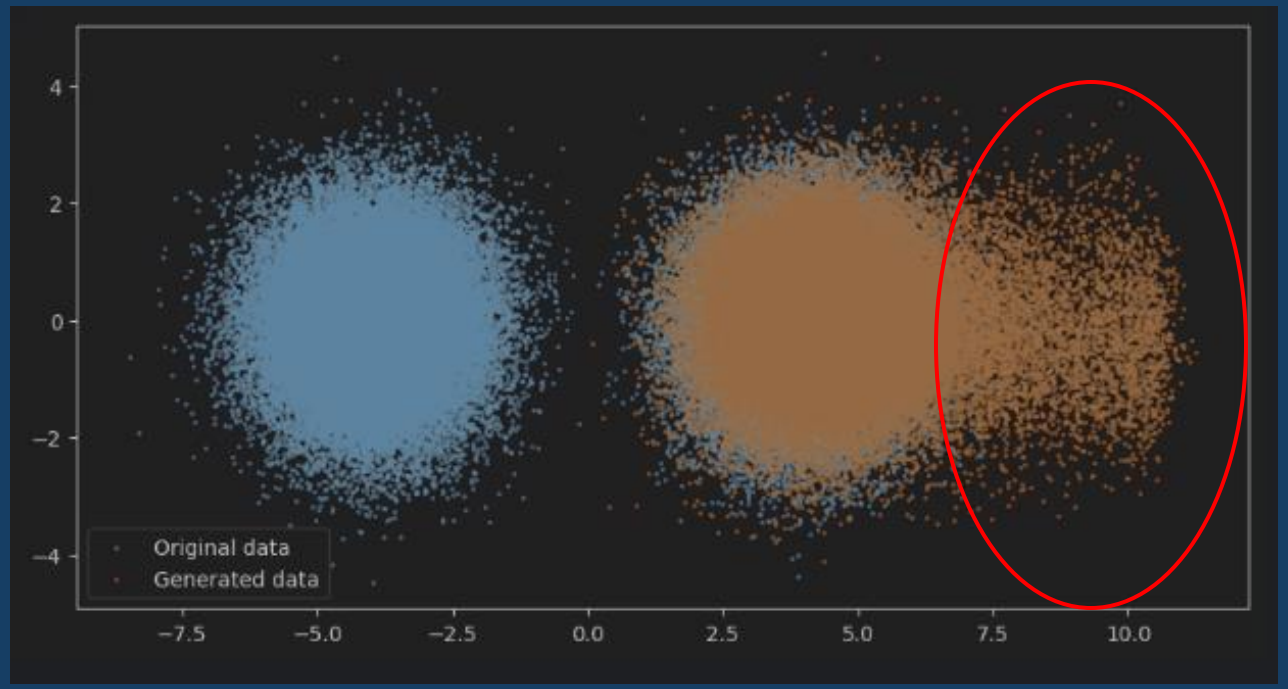
Guided generation targeting right distribution



Experiment Results

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z \quad + \quad g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term **guidance term**



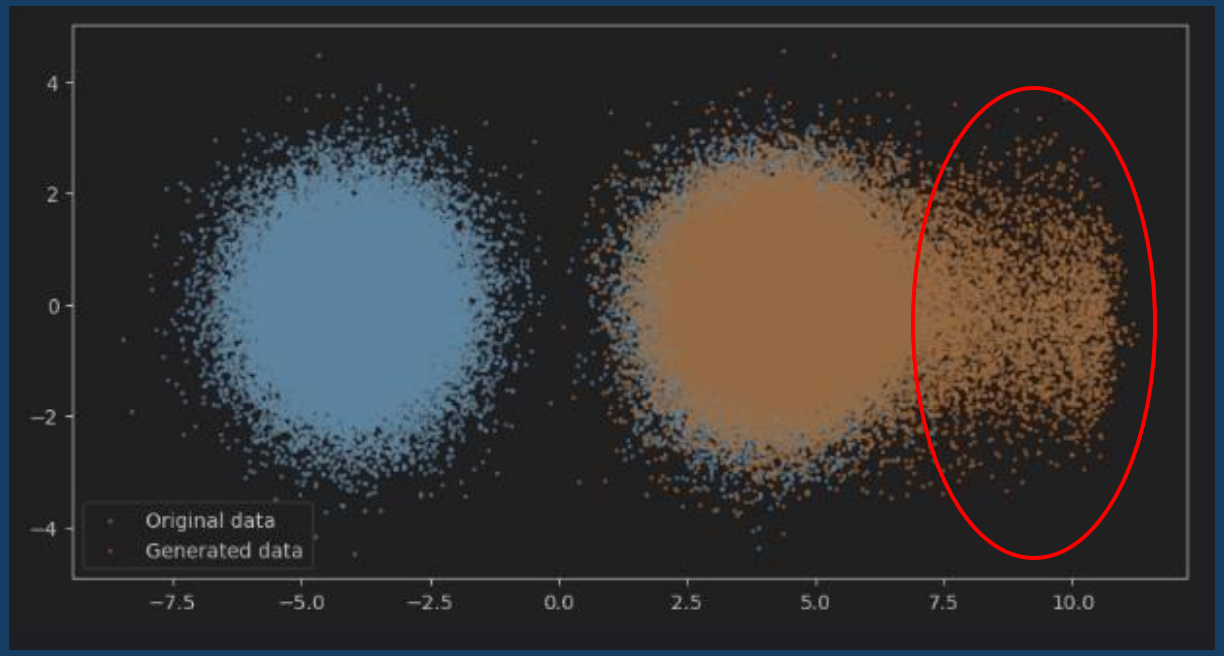
DDPM	Guidance scale g = 0	Guidance scale g = 1
Coverage (real→gen)	0.821	0.959 (+ 16.6%)
Coverage (gen→real)	0.872	0.859 (- 0.01%)
MMD^2	0.053	0.015 (- 71%)

96% real data are covered generated data
86 % generated data fall into real distribution
Even lower distribution difference

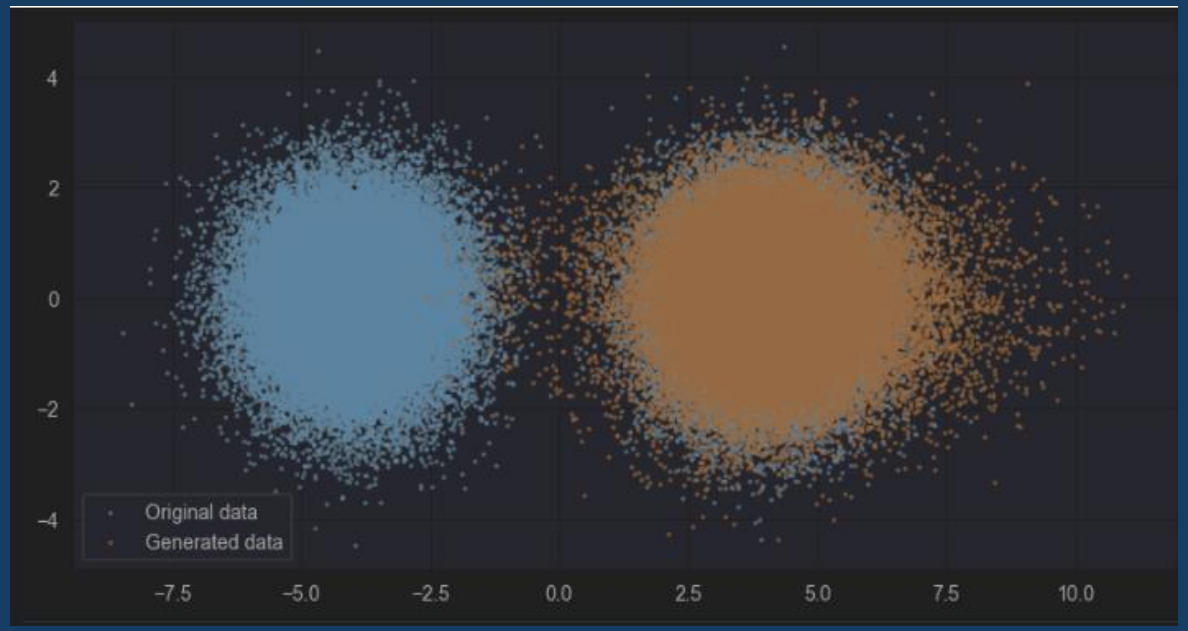
Experiment Results

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z + g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term | guidance term



Guidance scale g = 1



Guidance scale g = 0.4

Experiment Results

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z \quad + \quad g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term **guidance term**

DDPM	Guidance scale g = 0	Guidance scale g = 1	Guidance scale g = 0.4
Coverage (real→gen)	0.821	0.959 (+ 16.6%)	0.965 (+ 17.5%)
Coverage (gen→real)	0.872	0.859 (- 0.01%)	0.920 (+ 5%)
MMD^2	0.053	0.015 (- 71%)	0.0029 (- 94%)

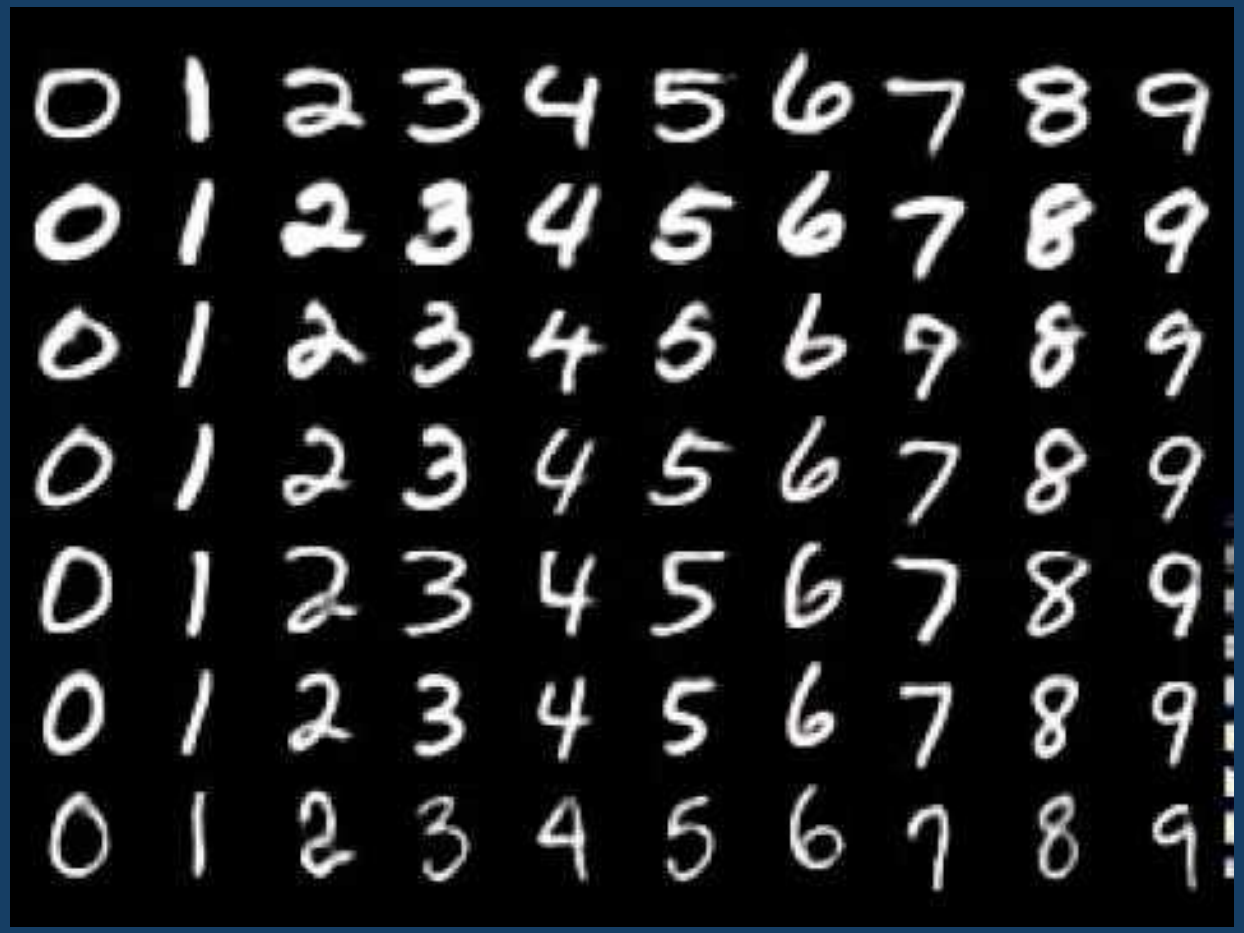
17% more real data are covered
5 % more gen data fall into real distribution

94% reduced distribution difference

Higher, better

Lower, better

Image dataset (MNIST numbers)



Experiment Setting

- **Dataset:** MNIST
- **Sampling:** Density ratio guidance

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z \quad + \quad g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term | guidance term

- **Quantitative Metrics**

Coverage (Gen→real) → **Precision:** how many generated data fall inside real distribution → Accuracy

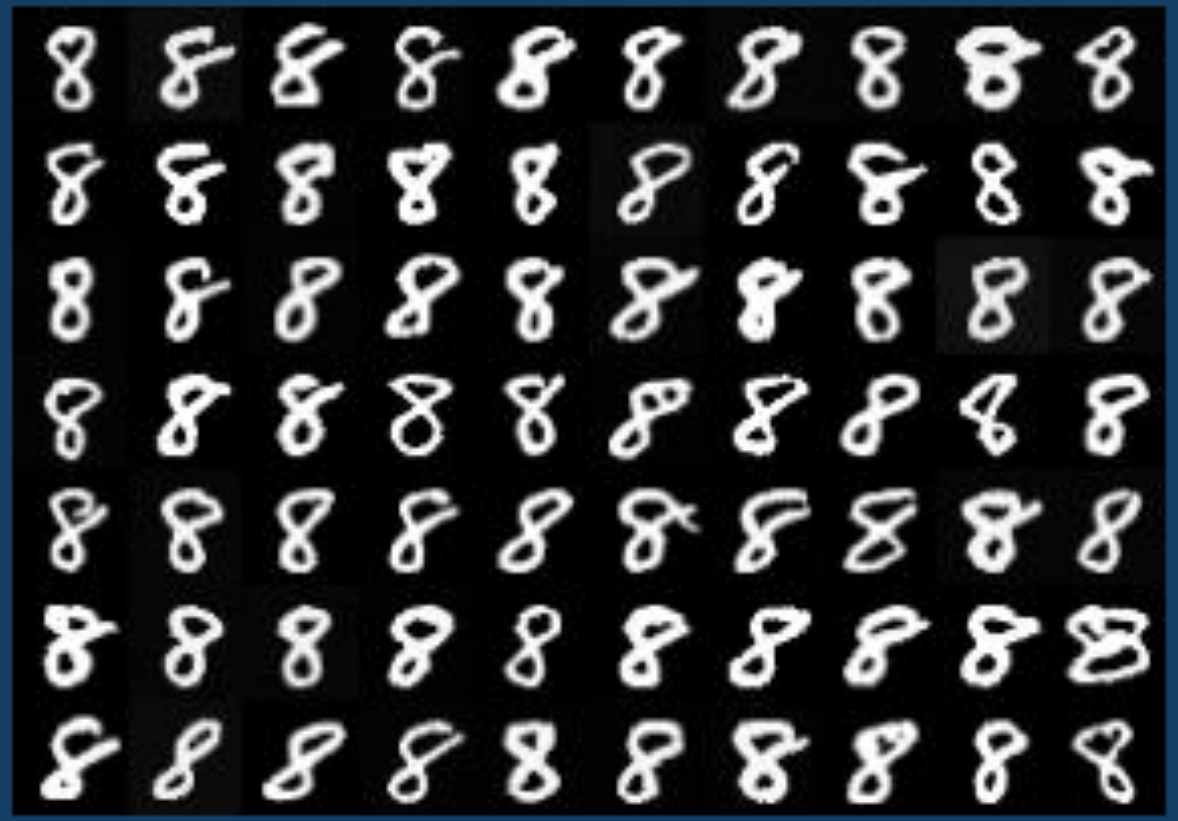
Coverage (Real→Gen) → **Recall:** how many real data are covered by generated data → Diversity

Experiment Results

Without Guidance



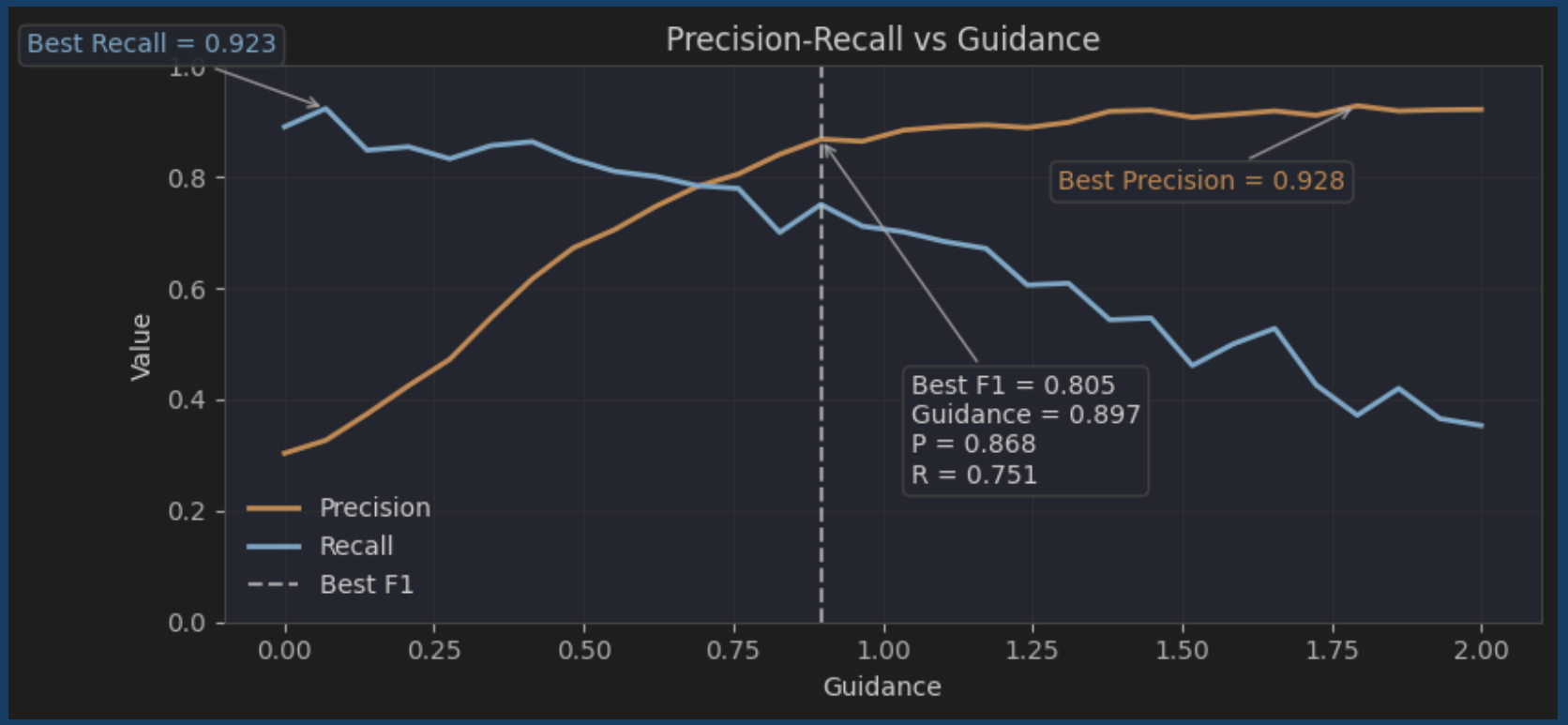
Guided generation targeting number 8



Experiment Results

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z \quad + \quad g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term **guidance term**



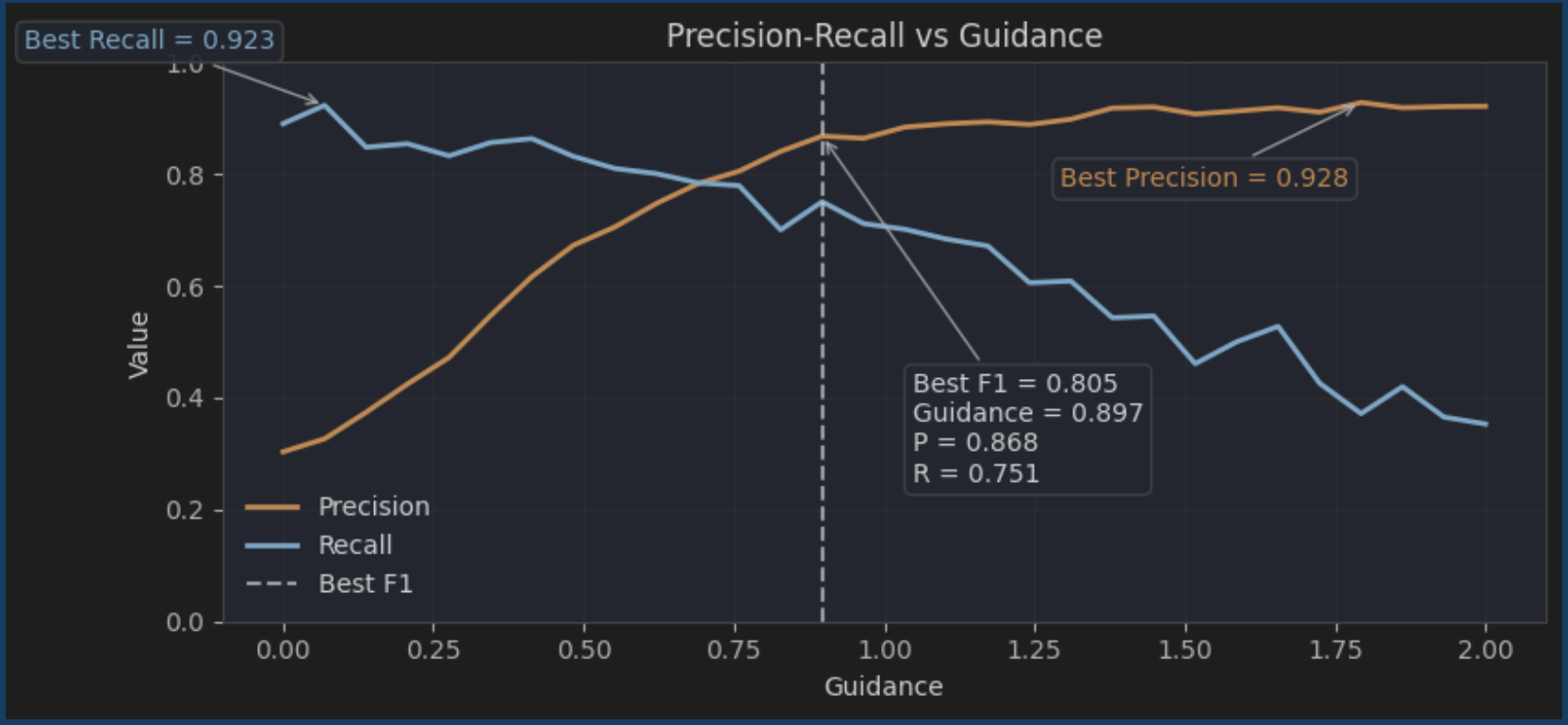
Precision	peaks at 0.93
Recall	peaks at 0.92
F1	peaks at 0.81

- As guidance g increases:
  - Precision increases
  - Recall decreases

Experiment Results

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z \quad + \quad g \cdot \beta_t \nabla_x \log\left(\frac{p}{q}\right)$$

DDPM sampling term **guidance term**



We choose the guidance strength that best balances the precision and recall

For g = 0.9

- 87% generated date fall inside the real distribution
- 75% real data are covered by generated data

Conclusion

Pros

- Controllability: Control sampling process using density ratio guidance
- Scalability: Performs on simple and complex dataset

Cons

- Flexibility: Sampling towards new distribution requires retraining the classifier
- Optimisation: You must optimise the guidance weight to balance the quality and diversity

Future

- Compare with other guidance techniques like classifier guidance
- Further the mathematical foundations, especially in optimisation.

Thank you for your time!

Any questions?